

RaceDay API Endpoint Plan

PROG6212 POE Part 1 | Student: Bono Nenguda

Purpose: This plan defines the RaceDay API before implementation. All routes begin with /api and use JSON request and response bodies unless stated otherwise.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new Organiser or Participant account and stores a hashed password.	None (public)	{ firstName, lastName, email, password, phoneNumber?, dateOfBirth?, roleName }	201 Created - user ID, name, email and role. 400 Bad Request - invalid or missing fields. 409 Conflict - email already registered.
POST	/api/auth/login	Authenticates a registered user and issues an access token.	None (public)	{ email, password }	200 OK - access token and user summary. 400 Bad Request - missing fields. 401 Unauthorized - invalid email or password.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the logged-in user's profile.	Any (logged in)	None	200 OK - user profile excluding PasswordHash. 401 Unauthorized - missing/invalid token. 404 Not Found - user no longer exists.
PUT	/api/users/me	Updates the logged-in user's editable profile details.	Any (logged in)	{ firstName, lastName, phoneNumber?, dateOfBirth? }	200 OK - updated profile. 400 Bad Request - invalid values. 401 Unauthorized. 404 Not Found.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists available events; optional query parameters may filter by type, date or location.	None (public)	None	200 OK - array of event summaries; an empty array when no events match. 400 Bad Request - invalid filter.
GET	/api/events/{eventId}	Returns one event with its organiser and category summary.	None (public)	None	200 OK - event details. 404 Not Found - event does not exist.
POST	/api/events	Creates a new running, walking or cycling event for the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, distance, eventType }	201 Created - created event. 400 Bad Request - invalid fields/type/distance. 401 Unauthorized. 403 Forbidden - not an Organiser.
PUT	/api/events/{eventId}	Updates an event owned by the logged-in Organiser.	Organiser (owner)	{ eventName, description, eventDate, location, distance, eventType }	200 OK - updated event. 400 Bad Request. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found.
DELETE	/api/events/{eventId}	Deletes an event owned by the logged-in Organiser when no dependent enrolments prevent deletion.	Organiser (owner)	None	204 No Content. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found. 409 Conflict - event has enrolments/results.
GET	/api/organisers/me/events	Lists all events created by the logged-in Organiser.	Organiser	None	200 OK - array of owned events. 401 Unauthorized. 403 Forbidden - not an Organiser.

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists the age and distance categories available for an event.	None (public)	None	200 OK - category array. 404 Not Found - event does not exist.
GET	/api/categories/{categoryId}	Returns one category and its parent event identifier.	None (public)	None	200 OK - category details. 404 Not Found.
POST	/api/events/{eventId}/categories	Adds a category to an event owned by the logged-in Organiser.	Organiser (event owner)	{ categoryName, categoryType, description? }	201 Created - category. 400 Bad Request - invalid type/fields. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found - event. 409 Conflict - duplicate category.
PUT	/api/categories/{categoryId}	Updates a category belonging to the Organiser's event.	Organiser (event owner)	{ categoryName, categoryType, description? }	200 OK - updated category. 400 Bad Request. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found. 409 Conflict - duplicate.
DELETE	/api/categories/{categoryId}	Deletes an unused category from an event owned by the Organiser.	Organiser (event owner)	None	204 No Content. 401 Unauthorized. 403 Forbidden. 404 Not Found. 409 Conflict - category is used by enrolments.

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the logged-in Participant in an event and selected category.	Participant	{ categoryId }	201 Created - enrolment with Pending status. 400 Bad Request - category not valid for event. 401 Unauthorized. 403 Forbidden. 404 Not Found - event/category. 409 Conflict - already enrolled.
GET	/api/enrolments/me	Lists all enrolments belonging to the logged-in Participant.	Participant	None	200 OK - array with event, category and status. 401 Unauthorized. 403 Forbidden - not a Participant.
GET	/api/enrolments/{enrolmentId}	Returns one enrolment to its Participant or the Organiser who owns the related event.	Participant (own) or Organiser (event owner)	None	200 OK - enrolment details. 401 Unauthorized. 403 Forbidden - no access. 404 Not Found.
GET	/api/events/{eventId}/enrolments	Lists Participants enrolled in an event owned by the logged-in Organiser.	Organiser (event owner)	None	200 OK - enrolment array. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found - event.
PUT	/api/enrolments/{enrolmentId}/status	Changes an enrolment status to Pending, Confirmed or Cancelled.	Organiser (event owner)	{ enrolmentStatus }	200 OK - updated enrolment. 400 Bad Request - invalid status. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found.
DELETE	/api/enrolments/{enrolmentId}	Cancels/removes the logged-in Participant's enrolment before a result exists.	Participant (own)	None	204 No Content. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found. 409 Conflict - result already recorded.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/results	Lists published results for an event.	None (public)	None	200 OK - results ordered by finishing position. 404 Not Found - event.
GET	/api/results/me	Returns the logged-in Participant's race-result history.	Participant	None	200 OK - results with event, date, category, finish time and position. 401 Unauthorized. 403 Forbidden.
GET	/api/results/{resultId}	Returns one result to its Participant or the Organiser who owns the event.	Participant (own) or Organiser (event owner)	None	200 OK - result details. 401 Unauthorized. 403 Forbidden - no access. 404 Not Found.
POST	/api/enrolments/{enrolmentId}/result	Captures finish time and finishing position for a confirmed enrolment.	Organiser (event owner)	{ finishTime, finishingPosition }	201 Created - result. 400 Bad Request - invalid time/position or enrolment not confirmed. 401 Unauthorized. 403 Forbidden. 404 Not Found. 409 Conflict - result already exists/position duplicate.
PUT	/api/results/{resultId}	Corrects finish time or position for a result owned by the Organiser's event.	Organiser (event owner)	{ finishTime, finishingPosition }	200 OK - updated result. 400 Bad Request. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found. 409 Conflict - position duplicate.
DELETE	/api/results/{resultId}	Removes an incorrectly captured result from the Organiser's event.	Organiser (event owner)	None	204 No Content. 401 Unauthorized. 403 Forbidden - not owner. 404 Not Found.